

Group Problem Set 27: Rotational + Translational Motion

1. What are the rotational inertias of an empty toilet paper roll ($R = 0.025\text{m}$, $L = 0.1\text{m}$, $m = 0.10\text{kg}$) and a full toilet paper roll ($R_1 = 0.1\text{m}$, $R_2 = 0.025\text{m}$, $L = 0.1\text{m}$, $m = 1\text{kg}$). The toilet paper roll is rolled down a ramp 1 m off the ground.
 - (a) Write the general conservation of energy equations for each item. Be sure to take into account the mass distributions in the equations for inertia.
 - (b) What is the difference in start height required for the toilet paper rolls to have the same velocity at the bottom of the ramp?
 - (c) What is the final velocity at the base of the ramp of each TP if they start from a height of 1 m?
 - (d) How far will each roll travel if they both continue to roll on a flat surface with a coefficient of friction of 0.1.
 - (e) Write the conservation of energy equations for each roll as it rolls across the surface.
 - (f) Looking only at the conservation of energy equations, which portion of each roll's trajectory is conserved?
 - (g) Why?

a) $mgh = \frac{1}{2} v^2 \left(m + \frac{I}{r^2} \right)$

$$I_{\text{hollow}} = \frac{1}{2} \lambda ((0.1)^2 + (0.025)^2) = 0.0053$$

$$I_{\text{solid}} = \frac{1}{2} (0.1) (0.025)^2 = 3.1 \times 10^{-5}$$

b) $h = \frac{v^2}{2g} \left(1 + \frac{I}{mr^2} \right)$

$\left(1 + \frac{I}{mr^2} \right)$ is the dependent term

$$h_{\text{hollow}} = 1 + \frac{3.1 \times 10^{-5}}{(0.1)(0.025)^2} = 1.5\text{m}$$

$$h_{\text{solid}} = 1 + \frac{0.0053}{(0.1)^2} = 1.53\text{m}$$

$$\Delta h = 0.03\text{m}$$

2. A ball of radius 0.1m and mass of 1 kg, starts from a height of 1 m, and rolls down a hill.
- (a) What is the velocity of the ball at the bottom of the ramp.
 - (b) The ball hits a stationary ball at the bottom of the ramp. The stationary ball has a mass of 2 kg and radius of 0.05 m. What is the final velocity of the second ball after impact if the first ball stops and the collision is completely elastic?
 - (c) What is the final height of the second ball if it rolls up a ramp after the collision?
 - (d) What would the mass need to be, if instead, the first ball hit a second ball with the same radius (0.1 m) and stopped as the second ball continued up the ramp to a final height of 2 m?

3. Two children (25 kg each, assume equal mass distribution of a bar, with length 1m) hold hands and spin in a circle. They have an angular velocity of 1 rad/s.
- (a) What will happen to their angular velocity if they pull each other into a tighter formation, reducing their effective length to 0.6 m?
 - (b) What is their angular velocity?
 - (c) What is their initial total kinetic energy?
 - (d) What is their final total kinetic energy?
 - (e) Is their motion conserved?

$$c) \quad V = \sqrt{\frac{2gh}{F}} = \sqrt{\frac{20}{F}}$$

$$F_n = 1.5 \quad V_n = 3.65 \text{ m/s}$$

$$F_s = 1.53 \quad V_s = 3.81 \text{ m/s}$$

$$d) \quad d = \frac{V^2}{2Mg} \quad M = 0.1$$

$$d_n = \frac{3.65^2}{2} = 6.66 \text{ m}$$

$$d_s = \frac{3.81^2}{2} = 7.27 \text{ m}$$

$$e) \quad \frac{1}{2} m v^2 + \frac{1}{2} I \left(\frac{v}{R} \right)^2 = Mgh$$

f) when on the ramp, energy is conserved. when on the flat, energy is not conserved.

g) the ramp is frictionless, flat has friction.